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Full Length Article

Bulk irradiation of ammonia for polarized target experiments

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ABSTRACT

We present the simulation, design, and experimental results of a straightforward method for irradiating solid ammonia (NH_3) and solid deuterated ammonia (ND_3) to optimize their use in dynamic nuclear polarization (DNP) for scattering experiments. A prerequisite for DNP is a certain density of electron spins in the target material, provided by the concentration of F-centers generated by electron beam irradiation. We have developed an approach that efficiently irradiates large quantities of NH_3 or ND_3, producing approximately 25 grams of target material per hour. A key feature of the method is its near-homogeneous dose delivery, ensuring consistent quality across the entire batch. The production of optimized target material is crucial for polarized fixed target scattering experiments at accelerator facilities worldwide.

1. Introduction

Irradiated ammonia, in the form of polarized proton (NH_3) or polarized neutron/deuteron (ND_3), is widely used in solid-state fixed target scattering experiments. This preference stems from ammonia's ability to achieve high polarization, its robust radiation resistance, and a moderate dilution factor—the ratio of free polarizable nucleons to the total number of nucleons in the molecule. Dynamic Nuclear Polarization (DNP) is essential for enhancing the polarization of these materials. DNP relies on the presence of a sufficient density of electron spins, for ammonia the electron spins are typically provided by paramagnetic centers created through electron beam irradiation.

When cooled below 195.4 K, ammonia transitions into various solid phases with distinct crystal structures. Under pressures below 0.3 GPa (approximately 3000 atm), ammonia crystallizes into a cubic unit cell where each NH_3 molecule forms weak hydrogen bonds with neighboring molecules. This structure enhances both the density and mechanical durability of ammonia, making it ideal for use as polarized target material compared to its glassy matrix state. However, the crystalline structure also contains inherent defects, such as surface irregularities, impurities, or contaminants, which must be minimized to produce high-quality target material.

Among the defects, F-centers – where an anionic vacancy in the crystal lattice is occupied by one or more unpaired electrons – are particularly significant for DNP due to their paramagnetic properties. These defects can absorb electromagnetic radiation, leading to crystal discoloration [1]. Efficiently creating and maintaining these F-centers is essential for achieving high polarization in ammonia.

The standard Crabb polarized target configuration requires irradiating ammonia at liquid argon temperatures to generate F-centers [2]. This setup includes a high-power microwave source oscillating around 140 GHz (the difference between Larmor frequency of free electrons and the nuclei of interest in a 5 T field) and a superconducting magnet. The ammonia [3], confined in a permeable Kel-F cup and cooled to 1 K, is polarized by the microwaves. Key parameters during irradiation include material temperature, radiation dose, and electron beam energy, which collectively determine the concentration of paramagnetic centers as well as paramagnetic type [4].

Warm irradiation (at approximately 87 K) is generally sufficient for NH_3, but achieving high polarization in ND_3 [5] requires both *warm* and *cold* irradiation. The *warm* irradiation preparation technique, initially developed in 1979 [6], involves irradiating ammonia under liquid argon to keep it solid and preserve the generated paramagnetic centers. Liquid argon is preferred over liquid nitrogen to avoid explosive compounds when open to the atmosphere [7].

The thermal equilibrium polarization of proton and deuteron are small around 1K and 5 T. However, electrons in these conditions can achieve over 99% polarization. Irradiating ammonia creates paramagnetic radicals, providing the needed unpaired electron spins. The DNP process utilizes these highly polarized electrons which combined with their fast relaxation rate provides the needed spin reservoir for pumping the nuclear polarization. The dipole–dipole interaction between

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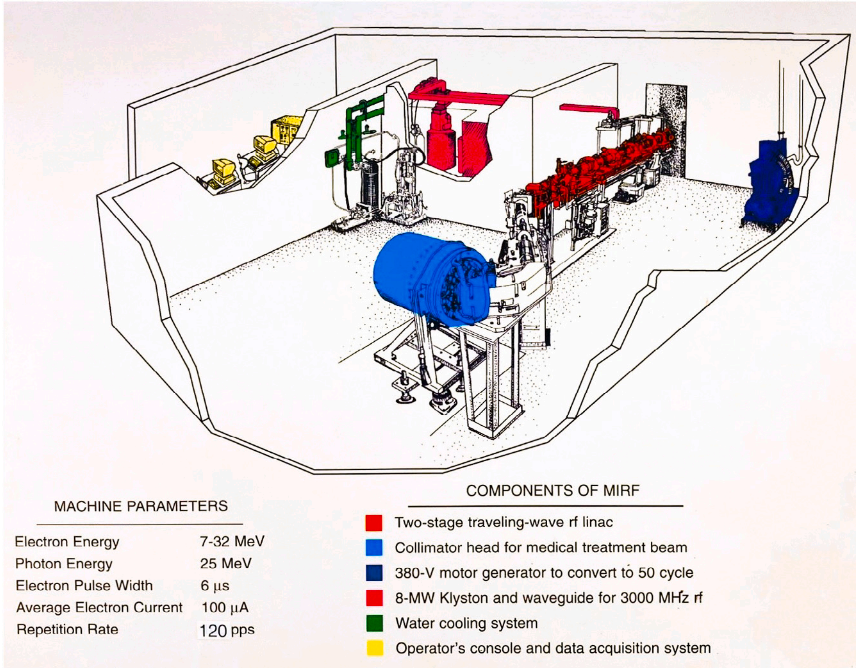

Fig. 1. The Medical-Industrial Radiation Facility (MIRF) at the National Institute of Standards and Technology (NIST).

nucleon and electron spins leads to hyperfine splitting, enabling the transfer of polarization from electron to nucleon spins driven by the microwave source.

By applying a suitable microwave signal, the desired spin state is populated by inducing RF-driven transitions between the unpaired electron spins (paramagnetic centers) and the nuclei spins (protons in NH_3 or deuterons in ND_3). The polarization mechanisms in ammonia vary depending on the isotopes involved [7]. Spin diffusion, the transport of polarization to nuclei beyond the immediate influence of paramagnetic centers, plays a crucial role in transmitting polarization to the bulk nuclei during the DNP process.

The present study uses the National Institute of Standards and Technology (NIST) high-intensity electron beam to irradiate ammonia, producing a dose of approximately 10^{17} e^{-}/cm^2. This method aims to optimize beam time efficiency and yield high-quality irradiated material.

This paper introduces a novel method for ammonia irradiation in polarized target experiments. The method employs a larger, rotating basket to hold the ammonia, eliminating the need for mid-irradiation rotation and increasing the target size for the beam. This design enhances the homogeneity of the dose distribution and maximizes the yield of irradiated material.

The remainder of this paper is organized as follows: Section 2 discusses the NIST facility. Section 3 describes the geometric setup for irradiation. Section 4 details the design of the new irradiation basket based on radiation dose simulations. Section 5 presents the outcomes of the irradiation process and tests for irradiation quality. Finally, Section 6 summarizes the benefits, drawbacks, and outlook for our design.

2. Irradiation facility

The Medical-Industrial Radiation Facility (MIRF) at NIST houses a traveling-wave linear accelerator (linac) with an energy range of 11 to 28 MeV and a maximum beam current of approximately 20 μA.

Originally a radiotherapy machine used for fifteen years at the Yale New Haven Radiation Oncology Center, the linac was donated to NIST in 1992. It has since been modified to include two high-intensity beamlines, including the zero-degree beam line used in this study. The linac operates with a pulse width of 6 μsec and a repetition frequency of 120 pulses per second.

At NIST, the MIRF, see Fig. 1, serves multiple applications. Initially, it developed calibration standards for cancer radiation treatment. Additional uses include radiation hardness testing of materials and devices, electron-beam curing, municipal waste sterilization, and materials modification studies, which is the focus of this application [8]. Fig. 2 shows a photograph of the MIRF linac at NIST.

For ammonia irradiation, the beam is set to 12.5 MeV and 10 μA, allowing rapid irradiation with minimal production of concerning isotopes. Using the traditional geometry (cylindrical target basket with axis parallel to the beam) [9], the fluence rate is approximately 6.8×10^{12} electrons/cm^2/s on the ammonia sample. To achieve the optimal fluence of 10^{16} to 10^{17} electrons/cm^2, irradiation is timed. The target material cup has a cross-section of 8 cm^2 and a length of about 6 cm. Halfway through the process, the cup is rotated 180 degrees to evenly distribute the dose, as there is higher dose deposition towards the beam in the elongated irradiation cup.

Fig. 3 shows the results of a Geant4 simulation of the dose delivered to the traditional target cup. The setup includes a beam flattener 400 mm downstream of the beam exit window, with the target material separated by 35 mm of polystyrene dewar wall and 25 mm of liquid argon. The total absorbed dose delivered to the ammonia is between 3 MGy and 5 MGy.

3. Updated irradiation setup and procedure

In the updated irradiation procedure, the target material is placed in a basket suspended within a scaffolding made of aluminum extrusions. A 1 rpm rotary motor is attached to the scaffolding, which rotates an aluminum rod holding the target basket in the beam path, 15 mm from the front edge of the dewar.

The target material and liquid argon are contained in a large foam dewar measuring 45 cm in length, 35 cm in height, and 32 cm in width, with 4 cm thick walls. Attached to the outside wall of the dewar, 40 cm downstream from the beam exit, are a zinc sulfide sheet (for visual beam monitoring) and a 1.5 mm thick aluminum beam widener. The leading edge of the target basket is positioned 1.5 cm from the interior wall of the dewar. Fig. 4 illustrates the new irradiation configuration.

Fig. 2. A photograph of the MIRF Linac. This portion of the accelerator is indicated in red in Fig. 1 as the Two-stage traveling-wave rf linac. This photo is taken downstream of the accelerator near the target area.

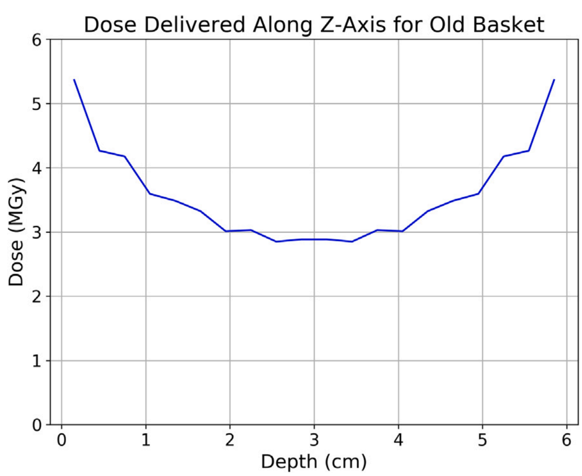

Fig. 3. Results of Geant4 modeling of the earlier elongated target cell irradiation setup. The dose shown assumes that the target cell was flipped halfway through the irradiation.

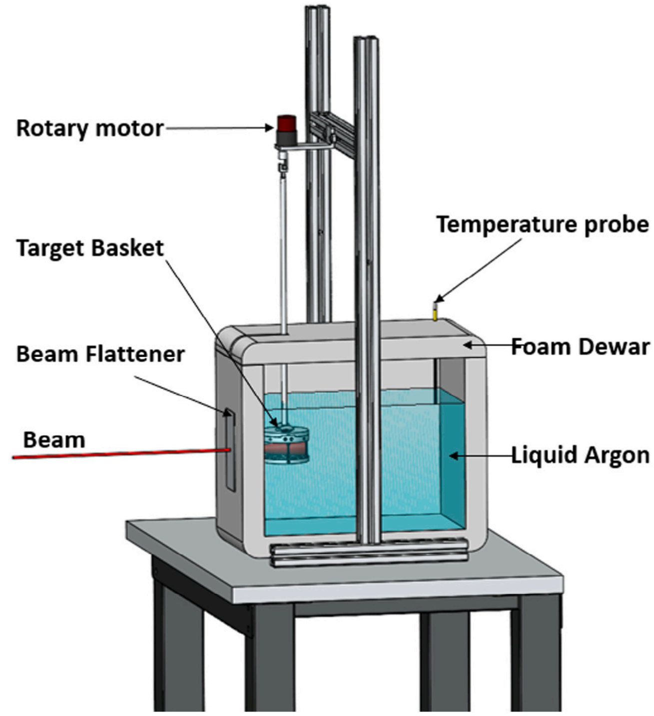

Fig. 4. The setup for irradiation under the new design. The beam exits the MIRF LINAC on the left and travels 400 mm to a beam widener. The beam then travels through 40 mm of foam and 15 mm of liquid argon before passing through the target basket. The basket has an inner diameter of 80 mm and a height of 30 mm.

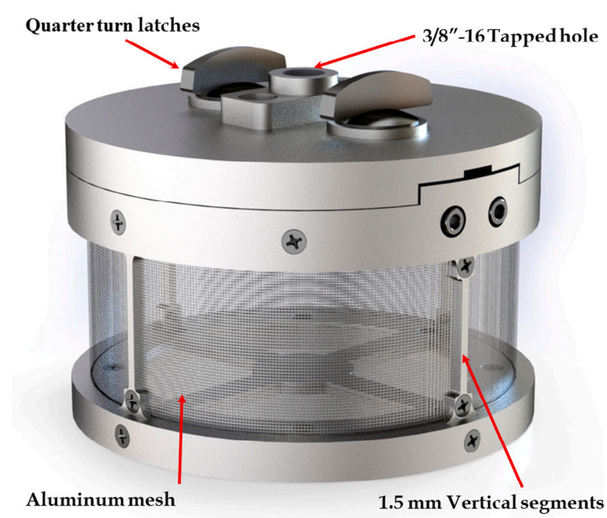

Fig. 5. A design for a rotary target basket to increase yields of irradiated ammonia.

To monitor the argon level in the dewar, a temperature probe is suspended approximately 2 cm above the target material. Temperature readings are monitored from the control room. When the temperature rises above 87 K, it indicates that the argon level has fallen just above the target material, signaling a need to pause irradiation and refill the argon. This occurs roughly once per hour, resulting in a beam stop of about 2 min.

4. Design of basket

To increase the yield and homogeneity of energy deposition, we designed a rotary basket (Fig. 5). The basket has an inner diameter of 8 cm and a beam window height of 3 cm. It is positioned in the beam path and rotated at 1 rpm to evenly distribute the dose. The outer frame is machined from a 6061 aluminum round tube, and both the bottom and outer wall are covered with an aluminum mesh to securely enclose the target material. The four vertical segments are machined to a width and thickness of 1.5 mm to minimize interaction with the beam. The basket lid is designed for easy opening and closing at low temperatures while securely holding the materials.

A 3/8’’-16 tapped hole on the lid connects to the rotary shaft, and two quarter-turn cam latches secure the lid to the basket frame. The rotating shaft, a 3/8’’ diameter, 18’’ long aluminum rod made of 6061 aluminum (Fig. 6), has a 3/8’’-16 coarse thread on one end for connecting to the basket lid. This design minimizes binding at low temperatures and ensures smooth engagement. A T-shaped stainless

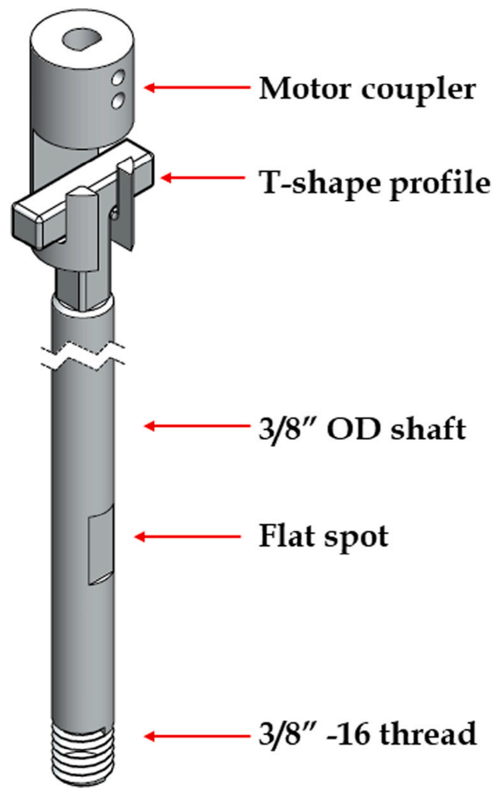

Fig. 6. The shaft assembly utilized with a coupler to connect with the motor, T-shaped profile, and 3/8”-16 thread to interface with the basket lid.

steel profile on the other end allows tool-free connection/disconnection of the shaft-basket assembly with the motor. The motor shaft connects to a coupler machined to accept the T-shaped end, securing the shaft inside.

A 12 V DC gear motor powers the shaft at a maximum speed of 1 rpm, with speed adjustments controlled by a pulse width modulation (PWM) variable driver. The motor assembly, made from aluminum T-slotted framing, can be positioned horizontally and vertically to ensure precise alignment with the beam.

The basket has a total interior volume of 151 cc, allowing for the irradiation of a large amount of ammonia in a single batch.

To determine the optimal basket size, we assessed the homogeneity of the dose received and the economic efficiency of material production. Other design considerations included ease of access to the ammonia and minimizing the amount of metal in the beam path to reduce activation.

To calculate the dose to the target, we modeled the electron beam assuming a Gaussian distribution. The intensity of a Gaussian beam is given by:

$$I(r,z) = I_0 \left(\frac{w_0}{w(z)}\right)^2 e^{\frac{-2r^2}{w(z)^2}}, \quad (1)$$

where r is the radial distance from the central beam axis, z is the axial distance from the beam waist, w_0 is the waist radius, and $w(z)$ is the beam radius at a given z. For our purposes, $w(z)$ is considered constant, giving:

$$I(r) = I_0 e^{\frac{-2r^2}{w^2}}. \quad (2)$$

The energy deposited per electron decreases as the beam travels through the argon and ammonia, modeled by the Landau distribution, with Geant4 simulation confirming this. The Landau distribution peaks approximately 1 cm into the material, before the electrons reach the

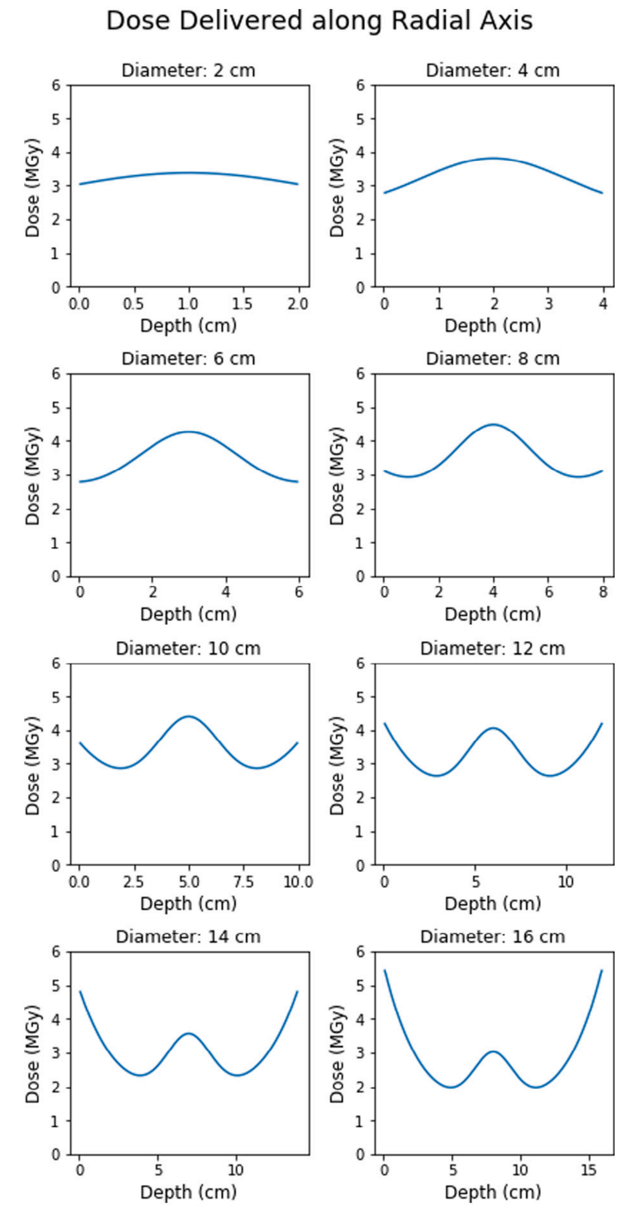

Fig. 7. Dose deposited in the ammonia for different rotary basket sizes as a function of depth, based on the model from Eq. (3) and times shown in Table 1.

ammonia [10]. Thus, the energy deposition is a combination of exponential decay in the z-direction and Gaussian in the radial direction:

$$\frac{dE}{dt} = Ae^{-\lambda z} e^{-\frac{r^2}{w^2}}, \quad (3)$$

where $e^{-\lambda z}$ represents exponential decay in the z-direction, and λ is the decay constant in units of cm^{-1}. The term $e^{-\frac{r^2}{w^2}}$ represents a Gaussian distribution in the radial direction (r) and $w^2 = 2\sigma^2$ with σ being the standard deviation of the Gaussian distribution such that w is the beam width that measures the radial spread of the beam's dose deposition. A is a constant representing the initial amplitude of the energy deposition which is in units of Watts. From our simulation, we obtained $\lambda = 0.432$ cm^{-1} and $w = 1.776$ cm. We calculated the energy deposited in the ammonia as a function of the radius, shown in Fig. 7.

We compared the energy deposition distribution by calculating the standard deviation of the normalized radial energy deposition, referred

Table 1
Non-uniformity, irradiation time (hours), volume (cc), and volumetric improvement in production over traditional baskets for different rotary basket diameters (cm), based on Eq. (3). The old basket has a non-uniformity of 0.21, a volume of 29 cc, and takes 80 min to irradiate. Baskets smaller than 14 cm improve uniformity, and baskets between 6 cm and 14 cm improve yield compared to the old design.

Diameter	Non-Uniformity	Irradiation time	Volume	Improvement
2	0.03	1.1	9.42	−60%
4	0.09	2.0	37.7	−16%
6	0.13	3.5	84.8	10%
8	0.12	5.4	151	26%
10	0.10	8.3	236	28%
12	0.15	13	339	20%
14	0.23	20	462	6%
16	0.33	30	603	−10%

to as non-uniformity. The non-uniformity is the standard deviation of the dose divided by the average dose (σ_D/D).

We also calculated the time needed to irradiate the basket to an average dose of 4 MGy, the basket volume, and the improvement over the older design in cubic centimeters irradiated per hour. Results for various basket sizes are shown in Table 1.

The improvement metric in Table 1 is based on volume. The analysis as a function of basket radius is used to determine the best geometry of the basket needed. For the case of a 12 cm diameter we get a volumetric yield 28.07 cc per hour results from the new method compared to the old method 21.75 cc per hour. The calculation is done as the new volume to old ratio (28.07/21.75 = 1.20), which gives a 20% improvement for this case. On the other hand, the 2 cm diameter has a volume of 9.42 cc, and takes 1.08 h (rounded to 1.1 h). That gives us 8.72 cc per hour. 8.72/21.75 = 0.4, which results in an -60% improvement, meaning old method would be better by 60%.

The old target cup has a non-uniformity of 0.21 and takes 80 min to irradiate 29 cc of ammonia. Baskets between 6 cm and 14 cm in diameter improve both yield and homogeneity. Based on yield and homogeneity, a basket diameter of around 10 cm is ideal. Our design uses an 8 cm diameter to complete irradiations more timely, considering the required preparation and cleanup.

A Geant4 simulation of the new basket design confirms improved homogeneity over the target length, allowing us to fine-tune the dose delivered to the ammonia. Fig. 8 shows the total dose delivered over a 5-h irradiation using a 12.5 MeV beam at 10 μA, as a function of radial depth into the basket. The simulation indicates that the dose delivered is more uniform than the simplistic model suggests, with all radial positions receiving within 10% of the average dose. The simulation indicates that over five hours, the target material receives a dose of 4-6$\times 10^{16}$ electrons per cm^2, within the range expected to achieve peak polarization.

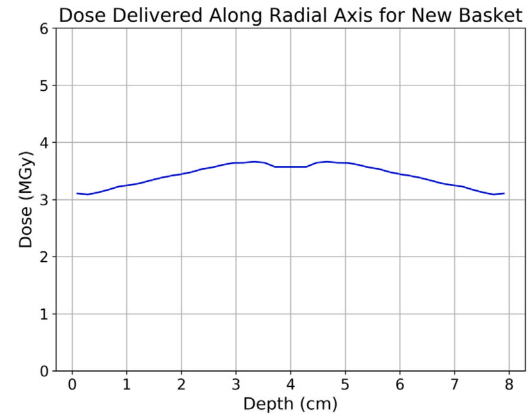

Fig. 8. Geant4 modeling of the received dose for the new irradiation setup along the radial axis. The Geant4 model shows a more homogeneous dose than the mathematical model, with a non-uniformity of 0.05. The dose is for a 5-h irradiation using a 12.5 MeV beam at 10 μA. This modeling assumes radial symmetry due to the constant rotation of the target basket.

Fig. 9. Ammonia that has been irradiated using the new basket design. The dark purple hue indicates that the desired dose has been achieved for high polarization.

5. Results

The new method has been used to irradiate NH_3 and ND_3 successfully and efficiently. NH_3 and ND_3 are kept separate and the irradiation is performed individually. In both cases after the irradiation, the entire basket of about 80 g of ammonia had a very even purple color throughout. This is an indication that the F-centers were evenly produced throughout the sample. A photograph after loading the NH_3 material into a storage bottle shows the promising results in Fig. 9. The purple hue is indicative of the appropriate scale of dose required to achieved high polarization for both NH_3 and ND_3. The ND_3 can only be further optimized and studied once it has an additional cold irradiation which can take place during the target commissioning of the scattering experiment. For NH_3 we can perform quality checks by polarizing it after it returns to the University of Virginia DNP lab.

The initial run for NH_3 produced approximately 77 g of optimized target material per day, which includes 2.5 h of setup and cleanup, and 5.5 h of irradiation at NIST. This compares to the old method, which allowed for approximately 45 g of target material to be produced per day, with 4 h of setup and cleanup and 4 h of irradiation. The improvement metric in Table 1 uses total volume increase which would be most relevant for solid slug or packed wafers where the packing fraction is 100%. For this study ammonia beads were used which results in a packing fraction of approximately 60%. We can use the packing fraction to convert the improvement metric into mass yield directly. Using irradiated mass to estimate the scale of improvement where $m = V\rho\eta$, where V is the volume, ρ is the density, and η is the packing fraction. The overall increase by mass in irradiated material is roughly 70% per-day.

The goal of this process was to generate optimized ammonia quickly and efficiently for a large amount of material, facilitating maximal polarization when used initially in scattering experiments. To check the quality of the process the material should be polarized using the DNP process. To do this a configuration similar to what is required in the setup for scattering experiments is used. This includes a 5 T

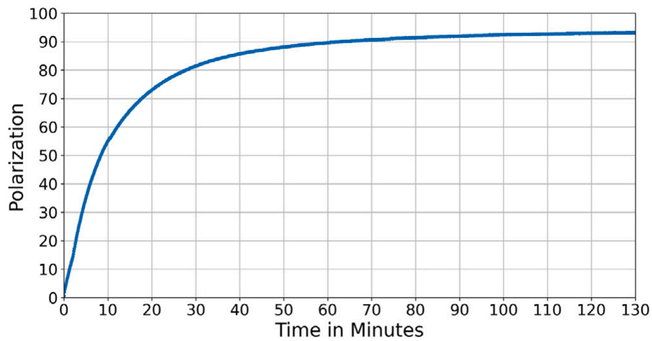

Fig. 10. The DNP ramp up polarization of the newly irradiated NH_3 at 5 T and 1 K indicating the approximate expected response.

superconducting magnet with field homogeneity on the scale of 10^{-4}, a high cooling power He-4 evaporation refrigerator with sufficient pumping capacity to cool the target material under the heat load from the Extended Interaction Oscillator (EIO) microwave generator at 1 K. A Phase-Matrix EIP frequency counter is used to measure the microwave frequency and a standard constant current continuous wave Q-meter based NMR system is used to measure the polarization [11]. Fig. 10 show a sample of 20 grams polarized over the course of several minutes using a microwave power of 270 mW. The polarization percentage is shown over time, measured in minutes. The ramp up rate and maximum polarization is in the expect rang for a optimal paramagnetic complex considering the microwaves used at 140.14 GHz indicating that the dose delivered to the target material is, on average, in the appropriate range.

The initial phase of the curve shows a steep increase in polarization percentage. This rapid rise occurs within the first 20 min. This is the critical quality check needed indicating that a significant number of spins align quickly once the irradiation process has induced sufficient paramagnetic centers. The observed polarization ramp-up is a direct consequence of the newly created color centers within the target material. The electron beam irradiation procedure worked well to evenly produce the necessary concentration and distribution of these centers. These results underscore the effectiveness of the irradiation process in creating the required amount of color centers necessary for maximal polarization, validating the new method and configuration used in this study.

6. Conclusions

This new design and procedure for irradiating ammonia target material for DNP experiments offer several key improvements over the previous setup. First, it reduces the time required to handle the target material, enhancing safety. Second, it improves the homogeneity of the dose delivered to the target material. Lastly, it increases the irradiation rate per hour, allowing for more target material to be produced within the limited beam time available.

The polarization ramp-up test effectively demonstrates the successful production of the desired paramagnetic complexes within the target material. The steep initial rise and subsequent plateau in polarization percentage confirm that the electron beam irradiation at NIST, using the new method, was able to achieve expected maximal polarization. These results indicate that the irradiation process effectively creates the required number of color centers necessary for maximal polarization in a large amount of target material, thus validating the production method and configuration used.

While this design for irradiating ammonia target material is a significant improvement, there are still areas for enhancement. Different geometries could potentially yield increased production or better homogeneity. Although this design minimizes radial and circumferential inhomogeneity, there is still some inhomogeneity over the height of the basket. Future designs could address this issue, allowing for more even irradiation and higher-quality target material for polarization experiments.

CRediT authorship contribution statement

Arthur Conover: Writing – original draft. **Vibodha Yasas Sri Bandara:** Writing – original draft. **Dustin Keller:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Fred Bateman:** Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dustin Keller reports financial support was provided by University of Virginia. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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